

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – Short Answer

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answer
Weightage:	30 % of CIA	Total Marks:	100 Marks
CO1 Statement:	Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
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Section / Batch:	B.tech C17 / 2024	Date:	24/07/2026

Code of Conduct

I Lalith Kumar [192421321] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Lalith

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

100%

Questions

- Q1.** A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.
- Q2.** A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. During capacity planning, an engineer needs to know how many channels a single cell will typically be allocated.
- Q3.** A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.
- Q4.** During a site audit, engineers note the base station radiates 80 W while a nearby handset transmits just 4 W. They want to express this power gap as a decibel-based power control ratio ($10 \cdot \log_{10}$ of the ratio).
- Q5.** A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.
- Q6.** An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?
- Q7.** A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?
- Q8.** Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?
- Q9.** A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

1. Total Spectrum = 45 MHz = 45,000 KHz

Full duplex voice channel need 2 simplex channel

$$= 2 \times 25 = 50 \text{ KHz}$$

$$\text{Channels} = \frac{45,000}{50} = 900$$

2. Channels per cell = Total channels / N

$$= 900 / 7$$

$$= 128$$

3. Total voice channels = Carriers x slots / carrier

$$= 100 \times 8$$

$$= 800$$

4. Power ratio = 80 / 4 = 20

$$\text{Ratio Pn dB} = 10 \times \log(20)$$

$$= 13.01 \text{ dB}$$

$$\approx 13 \text{ dB}$$

5. C/I (dB) = signal (dBm) - Interference (dBm)

$$= -60 - (-92)$$

$$= 32 \text{ dB}$$

6. Total channels = 42,000 KHz / 300 KHz

$$= 140$$

$$\text{Channels per cell} = 140 / 7 = 20$$

7. Received power $\propto 1/d^n$

Doubling distance with $n=4$ reduces power by $2^4 = 16 \times$

$$\text{New power} = 64 / 16 = 4 \text{ W}$$

8. With $n=2$, doubling radius requires power increase by

$$2^2 = 4 \times$$

9. Total channels in cluster = cluster size x channels / cell

$$= 4 \times 15 = 60.$$

10. $V = 100 \text{ km/h} = 27.78 \text{ m/s}$. Doppler shift $= V \times f / c$

$$V \times f / c = 27.78 \times 900 \times 10^6 / 3 \times 10^8$$

$$\approx 83.3 \text{ Hz}$$

$$\approx 83 \text{ Hz}$$

11. slot duration = Frame time / slots

$$= 5 \text{ ms} / 8$$

$$= 0.625 \text{ ms}$$

12. carriers = Total spectrum / spacing

$$= \frac{24000}{200}$$

$$= 120.$$

13. TX power in dBm $= 10 \log_{10} (8000 \text{ mW})$

$$\approx 39.03 \text{ dBm}$$

Received power $= 39.03 - 33$

$$= 6.03 \text{ dBm}$$

$$\approx 6 \text{ dBm}$$

14. using long distance path loss model referenced to 1m
Scaling the distance to 2.5 km give an approximate
free space path loss of $\approx 123 \text{ dB}$

15. Total Capacity = calls/cells $\times$ cells

$$= 250 \times 6.$$

$$= 1500.$$

16. Active calls supported = arrival rate $\times$ avg duration

$$= 4000 \text{ calls/sec} \times 180 \text{ sec}$$

$$= 720,000.$$

17. Total bandwidth = channels $\times$ bandwidth / channel

$$= 150 \times 25$$

$$= 3750 \text{ KHz}$$

$$= 3.75 \text{ MHz}$$

No. of guard bands between 30 bands = 29.

$$\text{Wasted bandwidth} = 29 \times 40 \text{ KHz}$$

$$= 1160 \text{ KHz}$$

$$= 1.16 \text{ MHz}$$

19. Speed = 90 km/h = 25 m/s · Cell diameter = 2 × 1.5 km

$$\text{Time per cell} = 3000 / 25$$

$$= 3000 \text{ m}$$

$$= 120 \text{ seconds}$$

$$\text{Handoffs/hour} = 3600 / 120$$

$$= 30$$

20. Total users = regional centers × base stations / center × users / base station

$$= 8 \times 40 \times 250$$

$$= 80,000$$

21. Approximate cell area $\approx R^2 = (0.4 \text{ km})^2 = 0.16 \text{ km}^2$

$$\text{No. of cells} = 400 / 0.16$$

$$= 2500$$

22. channels = Total spectrum / channel width

$$= 6000 / 30$$

$$= 200$$

23. Total calls - minutes = 1200 × 4 × 2.5

$$= 12,000 \text{ min/hr}$$

$$\text{Traffic in Erlangs} = 12,000 / 60$$

$$= 200$$

24. Users = Total bandwidth / bandwidth per user

$$= 40,000 \text{ KHz} / 400$$

$$= 100$$

25. Total Capacity = Users / cell × cells

$$= 100 \times 1200$$

$$= 120,000$$

26. Paging operations = Total cells / cells per operation

$$= 120,000 / 4 = 30,000$$

$$27. \text{Offered traffic} = \text{arrival rate} \times \text{avg duration}$$

$$= 40,000 \times 3 = 120,000 \text{ Erlangs}$$

$$\text{Excess over capacity} = 120,000 - 100,000$$

$$= 20,000$$

$$\text{Fraction blocked} = \frac{20,000}{120,000} \approx 0.17$$

$$28. \text{Roaming users} = 15\% \times 600,000$$

$$= 90,000$$

$$29. \text{Utilization} = \text{Peak traffic} / \text{Processing rate}$$

$$= 9,000 / 12,000$$

$$= 0.75$$

$$= 75\%$$

$$30. \text{Total channels} = \text{Bandwidth} / \text{channel width}$$

$$= 48,000 / 200$$

$$= 240$$

$$\text{Channels per cell} = 240 / 4 = 60$$

$$31. \text{channels} = \text{Total Spectrum} / \text{channel width}$$

$$= 30,000 / 200$$

$$= 150$$

$$32. \text{Successful handoffs} = 99\% \times 2500$$

$$= 2475$$

$$33. \text{Tx power in dBm} = 10 \log_{10} (12,000 \text{ mW})$$

$$\approx 40.79 \text{ dBm}$$

$$\text{Received power} = 40.79 - 45$$

$$\approx -4.21 \text{ dBm}$$

$$\approx -4 \text{ dBm}$$

$$34. \text{Total Capacity} = \text{users/cell} \times \text{cells}$$

$$= 900 \times 7$$

$$= 6300$$

$$35. \text{Slot duration} = \text{Frame time} / \text{slots}$$

$$= 4.8 \text{ ms} / 8$$

$$= 0.6 \text{ ms}$$

$$36. V = 150 \text{ km/h} = 41.67 \text{ m/s}$$

$$\text{Doppler shift} = v \times f / c = 41.67 \times 1.8 \times 10^9 / 3 \times 10^8$$

$$= 250 \text{ Hz}$$

$$37. \text{ offered traffic} = 70000 \times 4 = 280000 \text{ Erlangs}$$

$$\text{Excess} = 280000 - 250000$$

$$= 30,000$$

$$\text{Fraction Blocked} = 30000 / 280000$$

$$\approx 0.11.$$

$$38. \text{ Total users in cluster} = \text{users/cell} \times \text{cells in cluster}$$

$$= 600 \times 9$$

$$= 5400$$

$$39. \text{ calls dropped/sec} = 0.4\% \times 60,000$$

$$= 240.$$

$$40. \text{ Path loss (dB)} = \text{Tx power (dBm)} - \text{Rx power (dBm)}$$

$$= 43 - (-87)$$

$$= 130 \text{ dB.}$$

$$41. \text{ Total Coverage} = \text{Coverage / station} \times \text{n.o. of station}$$

$$= 2.5 \times 150$$

$$= 375 \text{ km}^2$$

$$42. \text{ Total channels} = 48,000 / 200$$

$$= 240$$

$$\text{channels per cell} = 240 / 6$$

$$= 40.$$

$$43. \text{ paging cycles} = \text{Total users} / \text{users per cycle}$$

$$= 150,000 / 6000$$

$$= 25.$$

$$44. \text{ Hand offs per call} = \text{Total users} / \text{users per cycle}$$

$$= 150000 / 6000$$

$$= 25.$$

$$44. \text{ Handoffs per call} = \text{call duration} / \text{handoff interval}$$

$$= 9 / 3$$

$$= 3.$$

$$45. \text{ SNR (dB)} = \text{Signal (dBm)} - \text{Noise (dBm)}$$

$$= -85 - (-113)$$

$$= 30 \text{ dB.}$$

$$46. \text{ Minimum bandwidth} = \text{users} \times \text{rate / user}$$

$$= 250 \times 150 \text{ kbps}$$

$$= 37,500 \text{ kbps}$$

$$= 37.5 \text{ Mbps.}$$

$$47. \text{ Total call - min/hr} = \text{users} \times \text{calls/hr} \times \text{avg duration}$$

$$= 1500 \times 4 \times 2.5$$

$$= 15,000 \text{ min}$$

$$\text{Traffic in Erlangs} = 15,000 / 60$$

$$= 250.$$

$$48. \text{ Total bandwidth} = \text{channels} \times \text{bandwidth / channel}$$

$$= 250 \times 12$$

$$= 3000 \text{ kHz}$$

$$= 3 \text{ MHz.}$$

$$49. \text{ Total throughput} = \text{connections} \times \text{rate / connection}$$

$$= 1500 \times 600 \text{ kbps}$$

$$= 900,000 \text{ kbps}$$

$$= 900 \text{ Mbps.}$$

$$50. \text{ Area per tower} = \pi R^2 = \pi \times 4^2 \approx 50.27 \text{ km}^2$$

$$\text{Towers needed} = 350 / 50.27$$

$$= 6.96$$

$$= 7.$$